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Which activity is MOST directly responsible for producing the final deliverable of a software project?

- A) Requirements analysis
 - B) Software design
 - C) Software construction
 - D) Software maintenance
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A developer writes code that works correctly now but fails after minor requirement changes. Which construction goal was MOST neglected?

- A) Easy to understand
 - B) Ready for change
 - C) Safe from bugs
 - D) Reuse
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Hiding implementation details while exposing simple interfaces is an example of:

- A) Modularization
- B) Verification

C) Abstraction

D) Integration

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Which practice MOST helps reduce mental load on developers when understanding code?

A) Clever algorithms

B) Increased cyclomatic complexity

C) Minimizing complexity

D) Extensive debugging

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Which statement BEST describes constructing for verification?

A) Writing test cases after coding

B) Making software fault-free

C) Building code so faults are easily detected

D) Eliminating the need for testing

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Using well-tested libraries instead of writing new code is an example of:

A) Construction for reuse

B) Construction with reuse

C) Modular design

D) Defensive programming

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Developing a generic logging module intended to be used in multiple projects demonstrates:

A) Construction with reuse

B) Code cloning

C) Construction for reuse

D) Integration testing

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Which type of standard MOST directly affects variable naming and code formatting?

A) Documentation standard

B) Testing standard

C) Coding standard

D) Process standard

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Which life cycle model treats construction mainly as a coding activity performed after design completion?

- A) Agile
 - B) Evolutionary prototyping
 - C) Iterative
 - D) Waterfall
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In which development approach are design, coding, and testing activities typically interleaved?

- A) Waterfall
 - B) Staged delivery
 - C) Agile
 - D) Linear sequential
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Counting the number of linearly independent paths in a program measures:

- A) Code size
 - B) Code stability
 - C) Code complexity
 - D) Code productivity
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Which metric BEST reflects the quality of software during construction?

- A) Lines of Code
 - B) Defect Density**
 - C) Build Time
 - D) Code Commits per Day
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Which metric would MOST help evaluate the efficiency of the debugging process?

- A) Unit test coverage
 - B) Cyclomatic complexity
 - C) Debugging time**
 - D) Defects per KLOC
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Visual programming languages are MOST limited by their difficulty in:

- A) Being learned
- B) Expressing complex logic**
- C) Integrating components
- D) Supporting reuse

1 5

Formal notation languages are MOST commonly used when:

- A) Rapid prototyping is needed
 - B) User interfaces are complex
 - C) Mathematical precision is required
 - D) Development speed is the priority
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Which practice helps prevent future security vulnerabilities during construction?

- A) Increasing code size
 - B) Avoiding input validation
 - C) Sanitizing inputs
 - D) Delaying testing
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Which testing activity is typically performed immediately after writing a single module?

- A) System testing
- B) Acceptance testing

C) Integration testing

D) Unit testing

1 8

Delaying the integration of all components until they are fully completed is known as:

A) Incremental integration

B) Continuous integration

C) Phased (Big Bang) integration

D) Modular integration

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Which integration approach provides earlier feedback and easier error localization?

A) Big Bang

B) Phased

C) Incremental

D) Parallel

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Which activity is LEAST likely to be considered part of software construction?

A) Debugging

B) Unit testing

C) Coding

D) Requirements elicitation